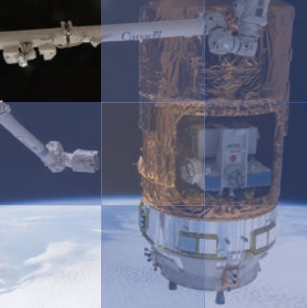


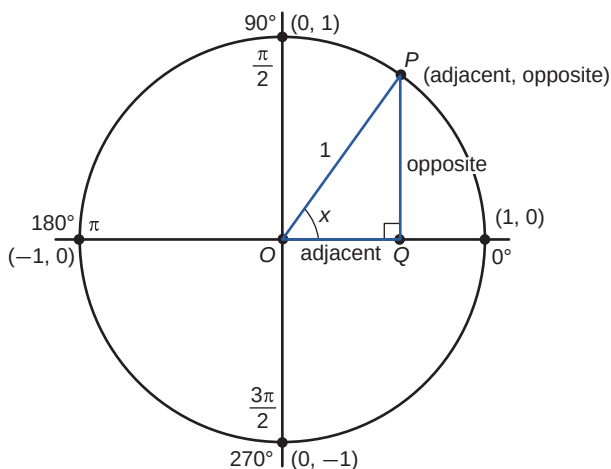


The Unit Circle

Trigonometric function values correspond to coordinates of points on the unit circle.

Identifying Points on the Unit Circle

Let P be a point located at the intersection of the terminal side of angle x and the unit circle. The length of the side of $\triangle POQ$ that is adjacent to x is the horizontal coordinate of P . The length of the side that is opposite x is the vertical coordinate of P .



Using the side lengths of $\triangle POQ$, you can establish the following ratios:

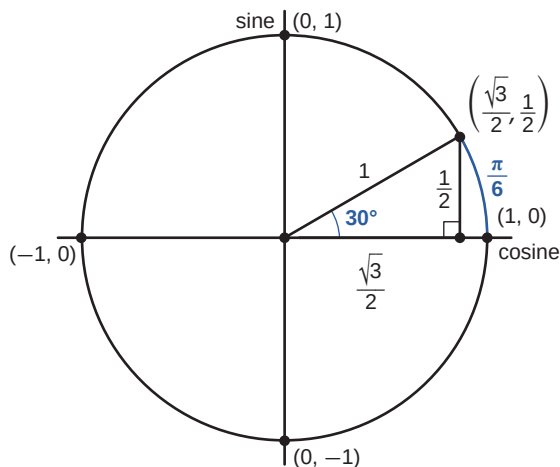
$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\text{adjacent}}{1} = \text{adjacent side length}$$

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\text{opposite}}{1} = \text{opposite side length}$$

For any point (a, b) on the unit circle that lies on the terminal side of an angle x in standard position, $a = \cos x$ and $b = \sin x$. For points on the unit circle, the horizontal axis is the cosine axis, and the vertical axis is the sine axis.

EXAMPLE 1

What is the point on the unit circle that lies on the terminal side of an angle in standard position with a measure of $\frac{\pi}{6}$?



SOLUTION

A $\frac{\pi}{6}$ radian angle measure is equivalent to 30° . The reference triangle is therefore a 30° - 60° - 90° special right triangle with hypotenuse 1, and legs with side lengths $\frac{\sqrt{3}}{2}$ and $\frac{1}{2}$.

The horizontal coordinate is the length of the horizontal leg.

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

The vertical coordinate is $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.

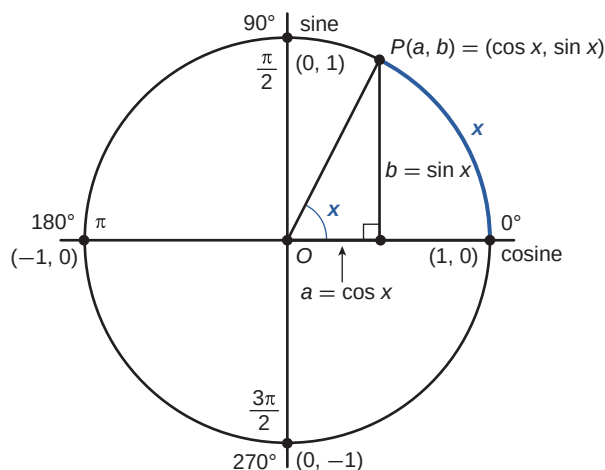
The coordinate pair of the point with a measure of $\frac{\pi}{6}$ is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. ■

Defining Trigonometric Functions

From the point $(1, 0)$ on the unit circle, move a distance of x units along the circle's circumference and stop at a point $P(a, b)$. Because P is on the unit circle, the real number x is the arc length from $(1, 0)$ to P and is equal to the radian measure of the angle that intercepts the arc. So a , the horizontal coordinate of P , is equal to $\cos x$ and b , the vertical coordinate of P , is equal to $\sin x$.

The movement x units from $(1, 0)$ along the unit circle takes you to a unique point (a, b) on the unit circle with coordinates $(\cos x, \sin x)$. Therefore, every real number x is paired with one and only one point $(\cos x, \sin x)$ on the graph of the unit circle. Because $a = \cos x$ and $b = \sin x$ are unique for any real number x , they can be defined as functions of x .

► **Remember** Distance is a length, so it is a real number.



Definitions

Let x be the radian measure of the angle that intercepts the unit circle at (a, b) . Then

$$\cos x = a \text{ and } \sin x = b$$

are trigonometric functions of x .

► **Think About It** If the movement from $(1, 0)$ to P is counterclockwise, then x is positive. If the movement is clockwise, then x is negative.

You have seen the tangent of x defined as the ratio of the side length opposite the angle x to the side length adjacent to x . With the unit circle, you can see that the tangent ratio for a radian angle measure x is $\tan x = \frac{\sin x}{\cos x}$.

► **Q&A**

Q What is the value of $\cos(\pi)$?

A -1

Definition

Let x be the radian measure of the angle that intercepts the unit circle at (a, b) . Then

$$\tan x = \frac{\sin x}{\cos x} = \frac{b}{a}$$

is a trigonometric function of x for all values of x that do not result in the cosine being zero.

Determining Trigonometric Function Values for Multiples of $\frac{\pi}{6}$, $\frac{\pi}{4}$, and $\frac{\pi}{3}$

You can find values of trigonometric functions for any angle measure or real number if you know the coordinates of the point on the unit circle that corresponds to that number. The point on the unit circle that corresponds

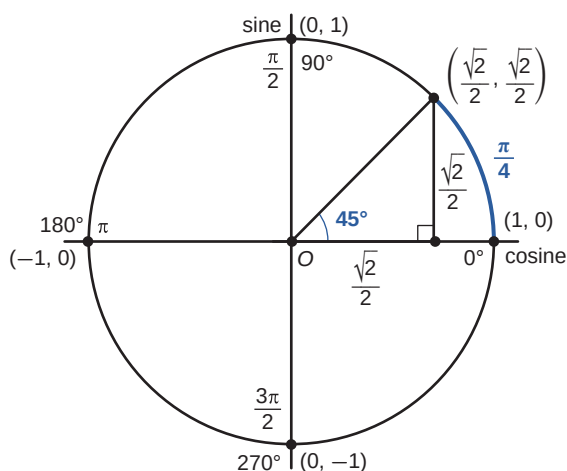
with $\frac{\pi}{6}$ is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, so $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ and $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. Use $\tan x = \frac{\sin x}{\cos x}$

$$\text{to get } \tan\left(\frac{\pi}{6}\right) = \frac{\sin\left(\frac{\pi}{6}\right)}{\cos\left(\frac{\pi}{6}\right)} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}}{3}.$$

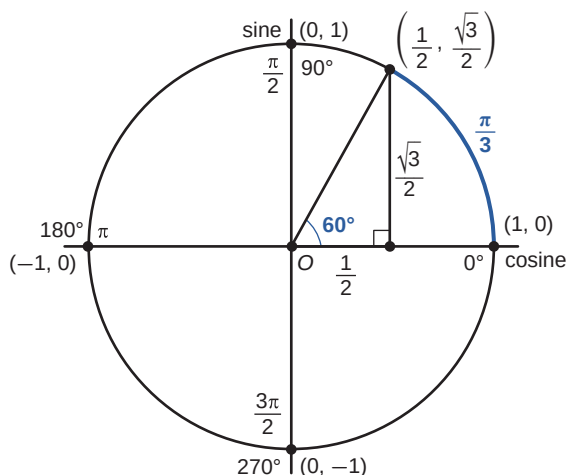
To determine the sine, cosine, and tangent of $\frac{\pi}{4}$, or 45° , draw a 45° - 45° - 90° triangle in Quadrant I of the unit circle.

The point that corresponds to $\frac{\pi}{4}$ is therefore $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, which means that

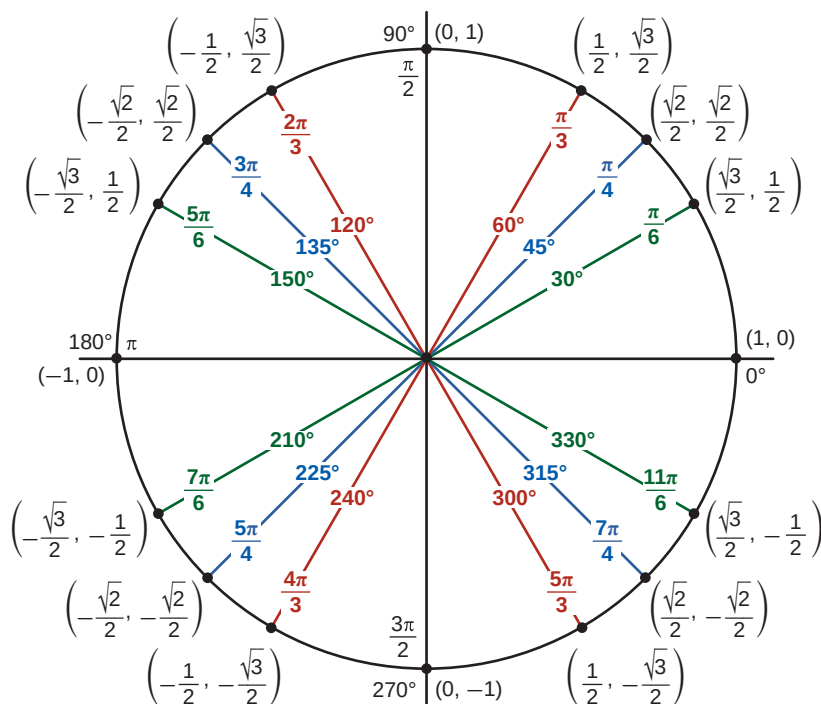
$$\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} \text{ and } \tan\left(\frac{\pi}{4}\right) = \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1.$$



To determine the trigonometric function values for $\frac{\pi}{3}$, use a 30° - 60° - 90° special right triangle, but use $\frac{\pi}{3}$, or 60° , as the reference angle.



When you move among the quadrants of the unit circle on the coordinate plane, the absolute values of the trigonometric functions of respective multiples of $\frac{\pi}{6}$, $\frac{\pi}{4}$, and $\frac{\pi}{3}$ are the same. The function values differ only by a sign.



EXAMPLE 2

Use the unit circle to find the exact value of $\sin\left(-\frac{2\pi}{3}\right)$.

► **Remember** Negative measures are generated by rotating clockwise.

SOLUTION

To determine the location of $-\frac{2\pi}{3}$, start at the positive horizontal axis. Then rotate $\frac{2\pi}{3}$ radians in the negative (clockwise) direction, which takes you into the third quadrant to the same point as $\frac{4\pi}{3}$. On the unit circle, $\frac{4\pi}{3}$ corresponds to the point $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$. So $\sin\left(-\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2}$. ■

► **Think About It** When locating the position of a measure using the unit circle, verify that the absolute value of the angle measure and its positive coterminal angle measure add to 2π . In this example,

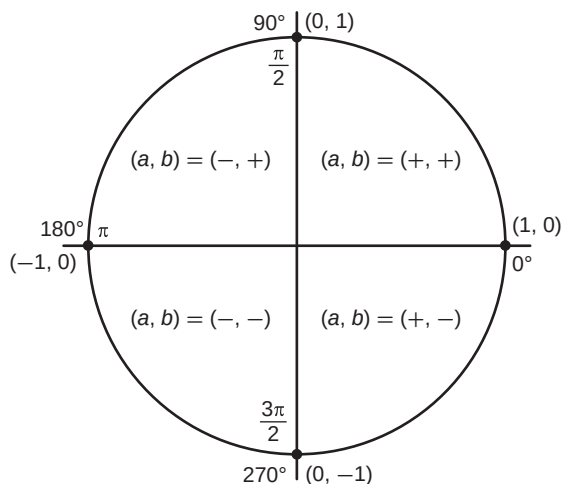
$$\frac{2\pi}{3} + \frac{4\pi}{3} = \frac{6\pi}{3} = 2\pi.$$

EXAMPLE 3

In which quadrant does x lie if $\sin x < 0$ and $\tan x > 0$?

SOLUTION

Look at the unit circle.



If x corresponds to the point (a, b) on the unit circle, then $\sin x = b$, which is negative in Quadrants III and IV. You can use $\tan x = \frac{b}{a}$ to see that $\tan x$ is positive when b and a have the same sign, which occurs in Quadrants I and III. The quadrant in which both conditions are true is Quadrant III. Therefore, x lies in Quadrant III. ■